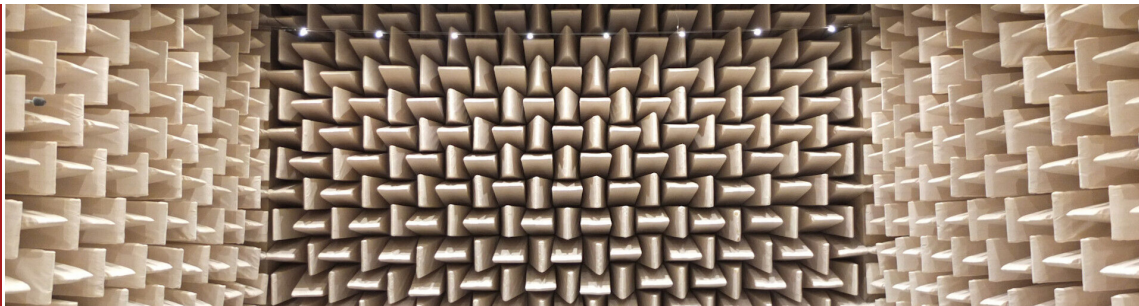




State-space Modelling of Head-related Transfer Functions

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Sales Pitch: Use state-space formulations

Advantages

- ▶ True MIMO modelling
- ▶ Numerical stability
- ▶ Physical insight
- ▶ Model order reduction
- ▶ Structure

Head-related transfer functions (HRTFs) are particularly well-suited

Disclaimer

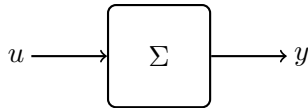
- ▶ I have no idea what I am talking about
- ▶ Math (sorry)
- ▶ Theory is simplified

Outline

- ▶ Why?
- ▶ What are state-space models?
- ▶ Application: FABIAN HRTF Database
- ▶ Theorems + Insights
- ▶ Example

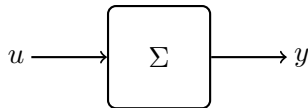
Linear Time-Invariant Systems (discrete-time)

- ▶ $u, y \in \ell_2(\mathbb{Z})$
- ▶ $\Sigma : \ell_2(\mathbb{Z}) \rightarrow \ell_2(\mathbb{Z})$
- ▶ $\Sigma(u) \mapsto y$



Linear Time-Invariant Systems (discrete-time)

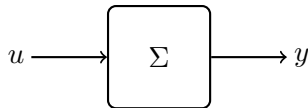
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- ▶ $\Sigma : \ell_2(\mathbb{Z}) \rightarrow \ell_2(\mathbb{Z})$
- ▶ $\Sigma(u) \mapsto y$



- ▶ **Linearity:** $\Sigma(\alpha u_1 + u_2) = \alpha \Sigma(u_1) + \Sigma(u_2), \quad \alpha \in \mathbb{R}$
- ▶ **Time-invariance:** $\Sigma \mathcal{S}_\tau(u) = \mathcal{S}_\tau \Sigma(u)$

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System

$$y_k = \Sigma(u)_k$$

Impulse Response

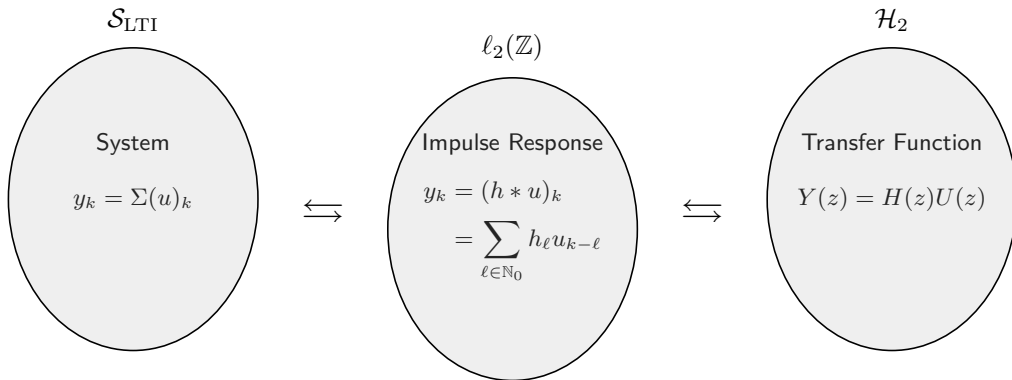
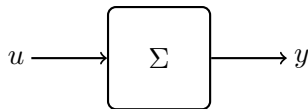
$$\begin{aligned} y_k &= (h * u)_k \\ &= \sum_{\ell \in \mathbb{N}_0} h_\ell u_{k-\ell} \end{aligned}$$

Transfer Function

$$Y(z) = H(z)U(z)$$

Linear Time-Invariant Systems (discrete-time)

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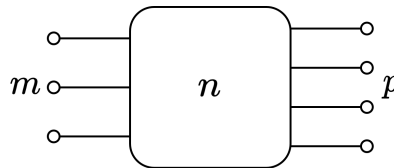
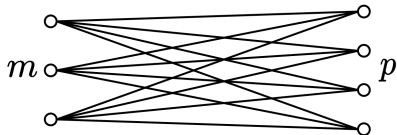


Problems

Numerical instabilities

$$H(z) = \frac{\sum_{k=0}^N b_k z^{-k}}{1 - \sum_{k=1}^M a_k z^{-k}}$$

MIMO complexity

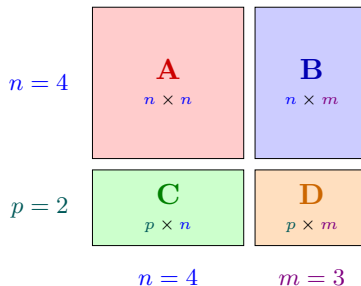


State-Space Realizations

Discrete-time state equations:

$$\begin{aligned}x_{k+1} &= \mathbf{A}x_k + \mathbf{B}u_k, \\y_k &= \mathbf{C}x_k + \mathbf{D}u_k.\end{aligned}$$

$$\mathbf{A} \in \mathbb{R}^{n \times n}, \mathbf{B} \in \mathbb{R}^{n \times m}, \mathbf{C} \in \mathbb{R}^{p \times n}, \mathbf{D} \in \mathbb{R}^{p \times m}$$



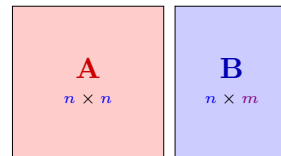
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$n = 4$



$p = 2$



$n = 4$

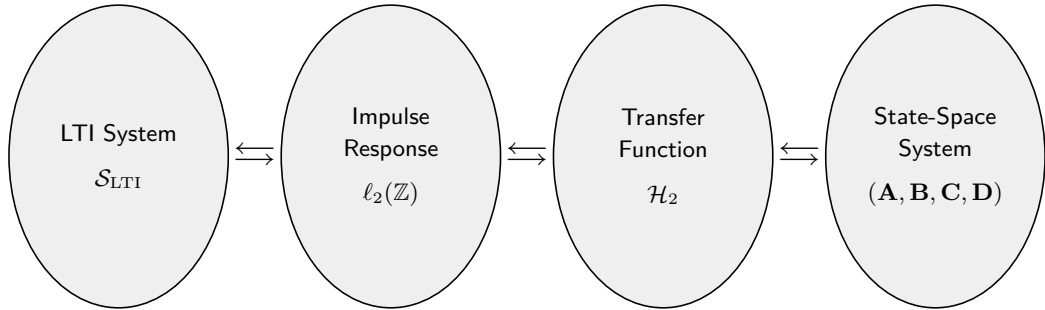
$m = 3$

► Impulse response:

$$h_k = \begin{cases} \mathbf{D}, & k = 0, \\ \mathbf{C}\mathbf{A}^{k-1}\mathbf{B}, & k \geq 1. \end{cases}$$

► Transfer function:

$$\mathbf{H}(z) = \mathbf{C}(z\mathbf{I}_n - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D}$$



What is the smallest possible n such that a state-space system has impulse response h ?

Hankel Matrix

MIMO impulse response $h_k \in \mathbb{R}^{p \times m}$

$$\mathcal{H} = \begin{bmatrix} h_1 & h_2 & h_3 & \cdots & h_s \\ h_2 & \ddots & & \ddots & h_{s+1} \\ h_3 & & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & & h_{2s-2} \\ h_s & h_{s+1} & \cdots & h_{2s-2} & h_{2s-1} \end{bmatrix} \in \mathbb{R}^{ps \times ms}$$

McMillan degree: $\text{rank}(\mathcal{H})$

How to obtain a state-space realization from impulse responses?

Eigensystem Realization Algorithm¹ (for dummies)

- ▶ Form Hankel matrix $\mathcal{H}$
- ▶ Compute (truncated) SVD of $\mathcal{H}$
- ▶ Solve a least-squares problem of size $(2s - 2)r^2$

¹J.-N. Juang and R. S. Pappa. *An Eigensystem Realization Algorithm for Modal Parameter Identification and Model Reduction*. *Journal of Guidance, Control, and Dynamics* 8.5 (Sept. 1985), pp. 620–627. DOI: [10.2514/3.20031](https://doi.org/10.2514/3.20031).

Head-Related Transfer Functions

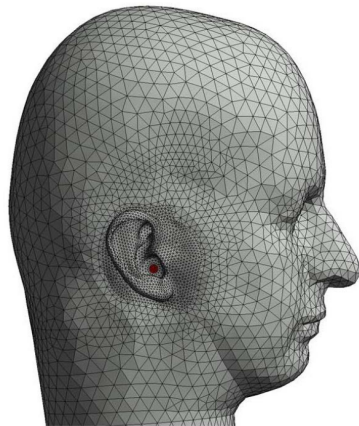


Image Source: F. Brinkmann et al. *A High Resolution and Full-Spherical Head-Related Transfer Function Database for Different Head-Above-Torso Orientations*. *Journal of the Audio Engineering Society* 65.10 (Oct. 2017), pp. 841–848. DOI: [10.17743/jaes.2017.0033](https://doi.org/10.17743/jaes.2017.0033)

- ▶ $m = 11950$ points on the sphere
- ▶ $p = 2$ ears

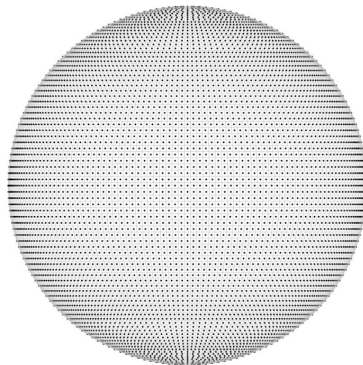


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- ▶ $m = 11950$ points on the sphere
- ▶ $p = 2$ ears
- ▶ 23900 impulse responses (measured and modeled)
- ▶ $f_s = 44\,100$ Hz
- ▶ $s = 256$

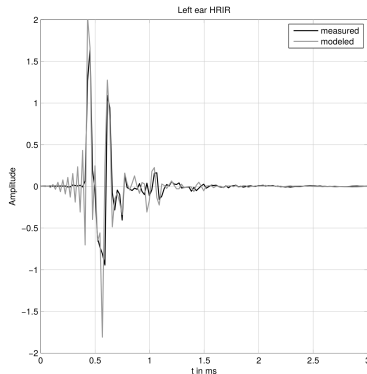


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$$\Rightarrow \mathcal{H} \in \mathbb{R}^{512 \times 3\,059\,200}$$

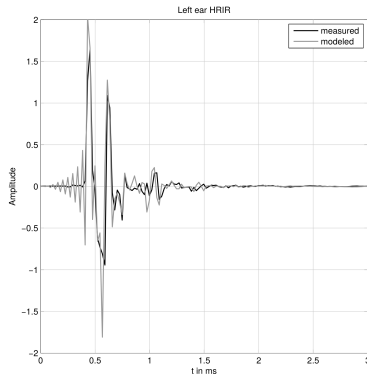


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Theorem. A MIMO-HRTF with $s \in \mathbb{N}$ samples has McMillan degree of at most $2s$.

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Bottleneck: SVD of $\mathcal{H} \in \mathbb{R}^{512 \times 3\,059\,200}$

Loss-free Virtual Input Projection

Tangential Eigensystem Realization Algorithm² Assume $h_k = 0$ for $k > s$.

$$\Theta_R = \begin{bmatrix} h_1 \\ \vdots \\ h_s \end{bmatrix} = \bar{\mathbf{U}} \bar{\mathbf{\Sigma}} \bar{\mathbf{V}}^\top \in \mathbb{R}^{ps \times m}, \quad q = \text{rank}(\Theta_R)$$

$$\mathbf{W}_2 = \bar{\mathbf{V}} \in \mathbb{R}^{m \times q}, \quad \mathbf{W}_2^\top \mathbf{W}_2 = \mathbf{I}_q, \quad \hat{h}_k = h_k \mathbf{W}_2$$

h_k	$\hat{h}_k$
$2 \times 11\,950$	2×512

²B. Kramer and S. Gugercin. *Tangential Interpolation-Based Eigensystem Realization Algorithm for MIMO Systems*. [Mathematical and Computer Modelling of Dynamical Systems](#) 22.4 (July 2016), pp. 282–306. DOI: [10.1080/13873954.2016.1198389](#).

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h_k	$\hat{h}_k$
$2 \times 11\,950$	2×512

$$\mathbf{P}_2 = \mathbf{W}_2 \mathbf{W}_2^\top, \quad h_k = h_k \mathbf{P}_2 = \hat{h}_k \mathbf{W}_2^\top, \quad k \geq 1,$$

$$\hat{\mathcal{H}} = \mathcal{H}(\mathbf{I}_s \otimes \mathbf{W}_2) \implies \sigma_i(\hat{\mathcal{H}}) = \sigma_i(\mathcal{H}), \quad i = 1, \dots, \text{rank}(\mathcal{H}).$$

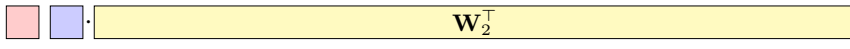
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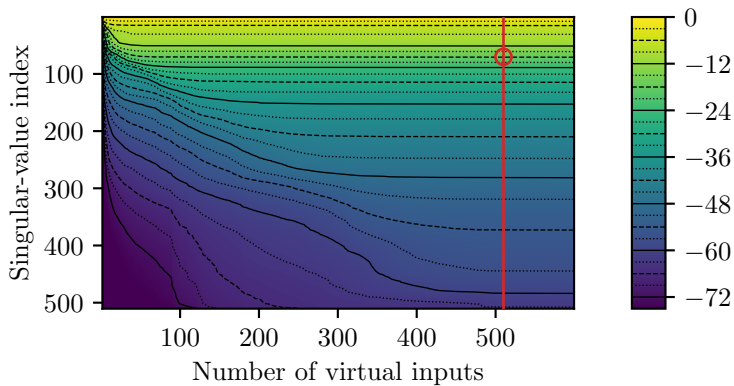
Loss-free Projected Eigensystem Realization Algorithm (for dummies)

- ▶ Calculate SVD of Θ_R and form $\mathbf{W}_2$
- ▶ Project impulse responses with $\mathbf{W}_2$
- ▶ Form **projected** Hankel matrix $\hat{\mathcal{H}}$
- ▶ Compute (truncated) SVD of $\hat{\mathcal{H}}$
- ▶ Solve a least-squares problem of size $(2s - 2)r^2$
- ▶ Backproject $\hat{\mathbf{B}}\mathbf{W}_2^\top = \mathbf{B}$

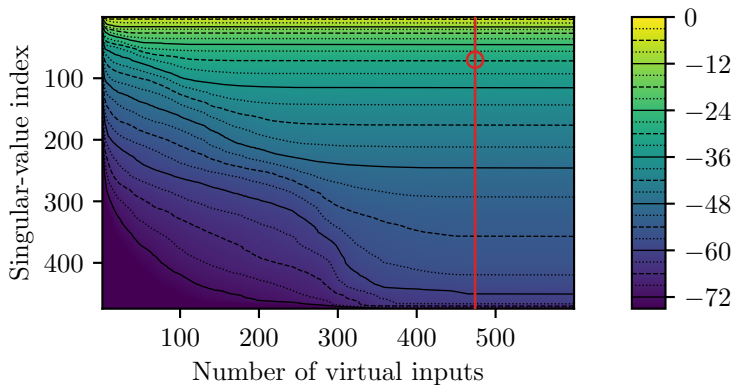
One SVD $512 \times 3\,059\,200 \Rightarrow$ two SVDs $512 \times 11\,950$ and $512 \times 131\,072$



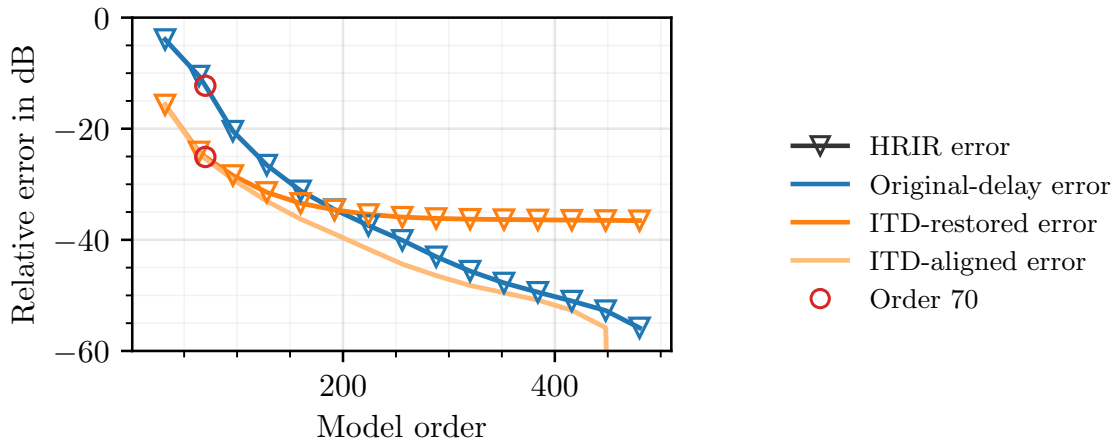
Hankel Singular Values



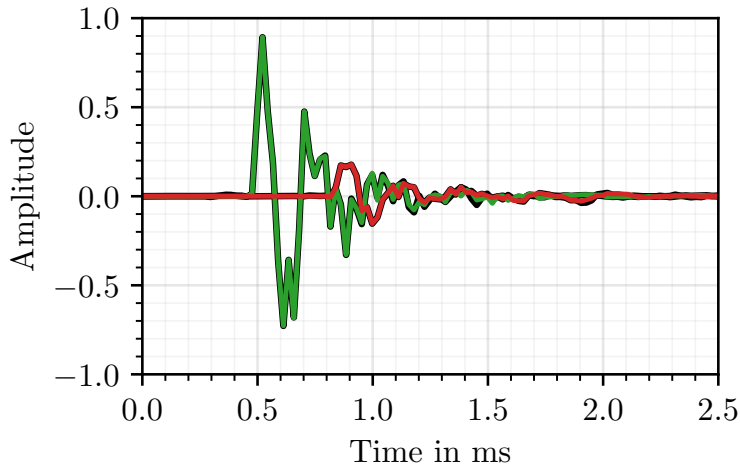
Hankel Singular Values: ITD removed



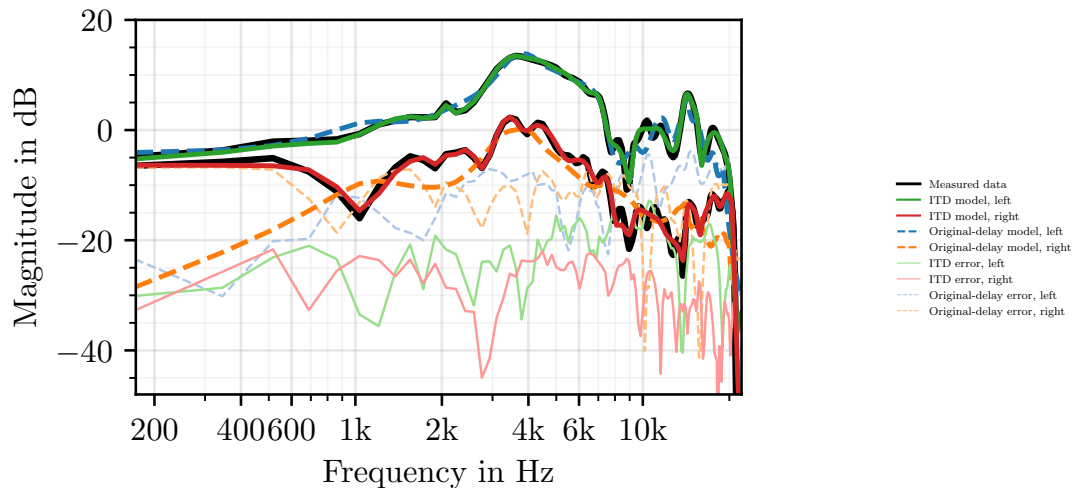
Relative Error



Impulse Responses



Transfer Function



References I



Brinkmann, F. et al. *A High Resolution and Full-Spherical Head-Related Transfer Function Database for Different Head-Above-Torso Orientations*. *Journal of the Audio Engineering Society* 65.10 (2017), pp. 841–848. DOI: [10.17743/jaes.2017.0033](https://doi.org/10.17743/jaes.2017.0033).



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github.com/artpelling/fa26-hrtf-ssm

State Transformations

Let $\mathbf{T} \in \mathbb{R}^{n \times n}$ be invertible and define

$$\begin{aligned}\tilde{x}_k &= \mathbf{T}^{-1}x_k. \\ \tilde{\mathbf{A}} &= \mathbf{T}^{-1}\mathbf{A}\mathbf{T}, \quad \tilde{\mathbf{B}} = \mathbf{T}^{-1}\mathbf{B}, \\ \tilde{\mathbf{C}} &= \mathbf{C}\mathbf{T}, \quad \tilde{\mathbf{D}} = \mathbf{D}.\end{aligned}$$

► Impulse response:

$$\tilde{h}_k = \tilde{\mathbf{C}}\tilde{\mathbf{A}}^{k-1}\tilde{\mathbf{B}}, \quad k \geq 1$$

► Transfer function:

$$\tilde{\mathbf{H}}(z) = \tilde{\mathbf{C}}(z\mathbf{I}_n - \tilde{\mathbf{A}})^{-1}\tilde{\mathbf{B}} + \tilde{\mathbf{D}}$$

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► Impulse response:

$$\begin{aligned}\tilde{h}_k &= \tilde{\mathbf{C}}\tilde{\mathbf{A}}^{k-1}\tilde{\mathbf{B}}, \quad k \geq 1 \\ &= \mathbf{C}\mathbf{T}(\mathbf{T}^{-1}\mathbf{A}\mathbf{T})^{k-1}\mathbf{T}^{-1}\mathbf{B} \\ &= \mathbf{C}\mathbf{A}^{k-1}\mathbf{B}\end{aligned}$$

► Transfer function:

$$\begin{aligned}\tilde{\mathbf{H}}(z) &= \tilde{\mathbf{C}}(z\mathbf{I}_n - \tilde{\mathbf{A}})^{-1}\tilde{\mathbf{B}} + \tilde{\mathbf{D}} \\ &= \mathbf{C}\mathbf{T}(z\mathbf{I}_n - \mathbf{T}^{-1}\mathbf{A}\mathbf{T})^{-1}\mathbf{T}^{-1}\mathbf{B} + \mathbf{D} \\ &= \mathbf{C}(z\mathbf{I}_n - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D}\end{aligned}$$